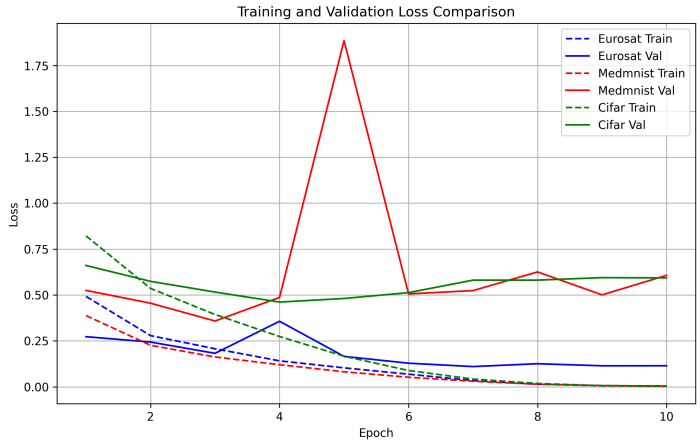


Figure 4: Comparison of training loss across different datasets. Models trained on EuroSAT and CIFAR-10 datasets demonstrate a steady decrease in loss, while the model trained on MedMNIST exhibits erratic loss curves, indicating instability during training due to domain mismatch.

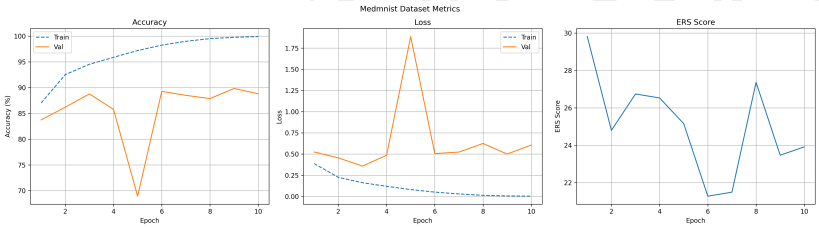


Figure 5: Performance metrics for the model trained on MedMNIST dataset. The erratic behavior in accuracy and loss indicates challenges in model convergence and generalization when applying the MedMNIST dataset to pest detection tasks.

**Comment**: This statement is incorrect again. It should say something like "...applying/using the MedMNIST dataset with the ImageNet-pretrained model." Also, the figure isn't mentioned in the main text.

The Environmental Robustness Score (ERS) is defined as:

$$\text{ERS} = \frac{\text{Accuracy under challenging conditions}}{\text{Accuracy under normal conditions}} \tag{1}$$

This metric quantifies the model's robustness to environmental changes by comparing its performance under augmented test sets to that under standard conditions.

The additional datasets used for multi-dataset training were:

- **EuroSAT**: A dataset consisting of 27,000 labeled Sentinel-2 satellite images covering 10 classes (**?**).
- **MedMNIST**: A collection of lightweight medical image datasets covering various tasks (**?**).
- **CIFAR-10**: A well-known dataset consisting of 60,000 32x32 color images in 10 classes (**?**).

For the multi-dataset training, we used a batch size of 64 to accommodate the increased data volume. Training was conducted for 30 epochs, and early stopping was applied if validation loss did not decrease for 5 consecutive epochs.

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